

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 7

End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objective of this experiment is to develop an end-to-end understanding of autoencoders and their variants for image representation, reconstruction, denoising and generative modeling.

The experiment begins with a fully connected autoencoder, progresses to a Convolutional Autoencoder (CAE), introduces image corruption for denoising, and finally develops a Variational Autoencoder (VAE). Students will visualize the learned latent representation, compare reconstruction quality using quantitative and qualitative measures, and explore the generative capability of the VAE.

The same image dataset and a controlled experimental protocol should be used throughout the experiment wherever possible.

2. Learning Outcomes

After completing this experiment, students will be able to:

- explain the encoder–latent–decoder structure of an autoencoder;
- prepare image data for reconstruction-based learning;
- implement a fully connected autoencoder;
- implement a Convolutional Autoencoder for image reconstruction;
- introduce controlled noise and build a denoising autoencoder;
- evaluate reconstruction using MSE, MAE and SSIM;
- visualize and interpret latent representations;
- explain the probabilistic latent representation of a VAE;
- implement the reparameterization trick;
- calculate and interpret reconstruction and KL-divergence losses;
- generate new images by sampling from a learned latent space; and
- compare deterministic and variational latent representations.

3. Dataset and Experimental Setup

Dataset: MNIST Handwritten Digit Dataset

MNIST contains grayscale images of handwritten digits from 0 to 9. Each image has spatial dimensions

$$28 \times 28 \times 1.$$

The dataset is particularly suitable for this experiment because the images are small, the classes have visually meaningful structure, and the same image format can be directly supplied to fully connected, convolutional and variational autoencoders.

The dataset can be loaded directly using Keras/TensorFlow:

https://www.tensorflow.org/api_docs/python/tf/keras/datasets/mnist

Recommended laboratory subset:

Use 10,000 training images and 2,000 test images for the main laboratory execution. The full MNIST dataset may be used if computational resources permit.

The labels are not required as reconstruction targets. The original image is the target:

$$x \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}$$

where x is the input image and $\hat{x}$ is its reconstruction.

4. Expected Input and Output

Expected Input

Each input image has shape

$$28 \times 28 \times 1.$$

For a batch of 32 images:

$$X \in \mathbb{R}^{32 \times 28 \times 28 \times 1}.$$

Pixel values should be converted from

$$[0, 255] \rightarrow [0, 1].$$

For a fully connected autoencoder, the image may be flattened:

$$28 \times 28 = 784.$$

Therefore,

$$x \in \mathbb{R}^{784}.$$

For a convolutional autoencoder, retain the spatial structure:

$$x \in \mathbb{R}^{28 \times 28 \times 1}.$$

Expected Output

The primary output is a reconstructed image:

$$\hat{x} \in \mathbb{R}^{28 \times 28 \times 1}.$$

Students must display:

Original image
Reconstructed image
Reconstruction error

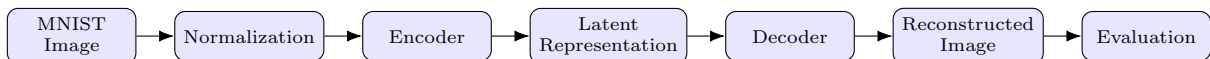
For the denoising model, display:

Original clean image
Noisy image
Denoised reconstruction

For the VAE, display newly generated samples obtained by sampling from the latent distribution.

5. Overall Experimental Pipeline

The complete experiment follows:



The experiment consists of four related studies:

1. Fully Connected Autoencoder
2. Convolutional Autoencoder
3. Denoising Convolutional Autoencoder
4. Variational Autoencoder

6. Exploring Autoencoders

An autoencoder learns to reproduce its input at the output.

The encoder maps the input to a lower-dimensional representation:

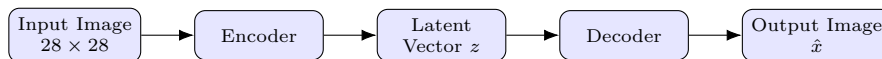
$$z = f_{\theta}(x),$$

and the decoder reconstructs the input:

$$\hat{x} = g_{\phi}(z).$$

The training objective is to minimize reconstruction error:

$$\mathcal{L}_{rec} = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|^2.$$



The latent vector is a compressed representation of the input.

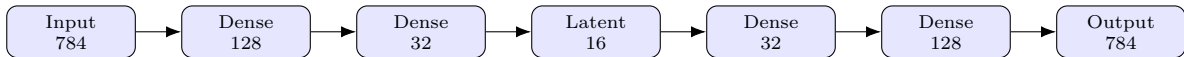
7. Fully Connected Autoencoder

Flatten each image from

$$28 \times 28 = 784$$

to a vector of length 784.

Use the following architecture:



Use ReLU activation in hidden layers and sigmoid activation in the output layer.

Suggested configuration:

Parameter	Value
Latent dimension	16
Optimizer	Adam
Learning rate	10^{-3}
Batch size	128
Epochs	20
Loss	Binary Cross-Entropy

8. Required Analysis of the Basic Autoencoder

Plot 1: Original versus Reconstructed Images



Figure 1: Original MNIST images and their autoencoder reconstructions.

Display at least 10 test images in a grid.

For every image, show:

Original | Reconstruction.

Students must comment on:

- preservation of digit shape;
- loss of fine details;
- blurred regions;
- differences between easy and difficult samples.

Inference: The overall shapes of the digits are retained well, with stroke direction and important topological features, such as loops and vertical strokes, preserved across the reconstructed samples. However, some fine details are lost; for example, thin strokes become thicker, as observed in the reconstruction of the digit 1. Blurring is most visible around loops and curved junctions, particularly in the digit 9 and in the ambiguous 5/6-like sample, where the loop becomes less distinct. Simple, mostly single-stroke digits such as 1 and 7 are reconstructed more accurately than digits containing loops or complex handwriting. This occurs because the 16-dimensional bottleneck must compress substantial visual information into a limited representation, causing intricate features to be smoothed or lost.

Plot 2: Training and Validation Reconstruction Loss

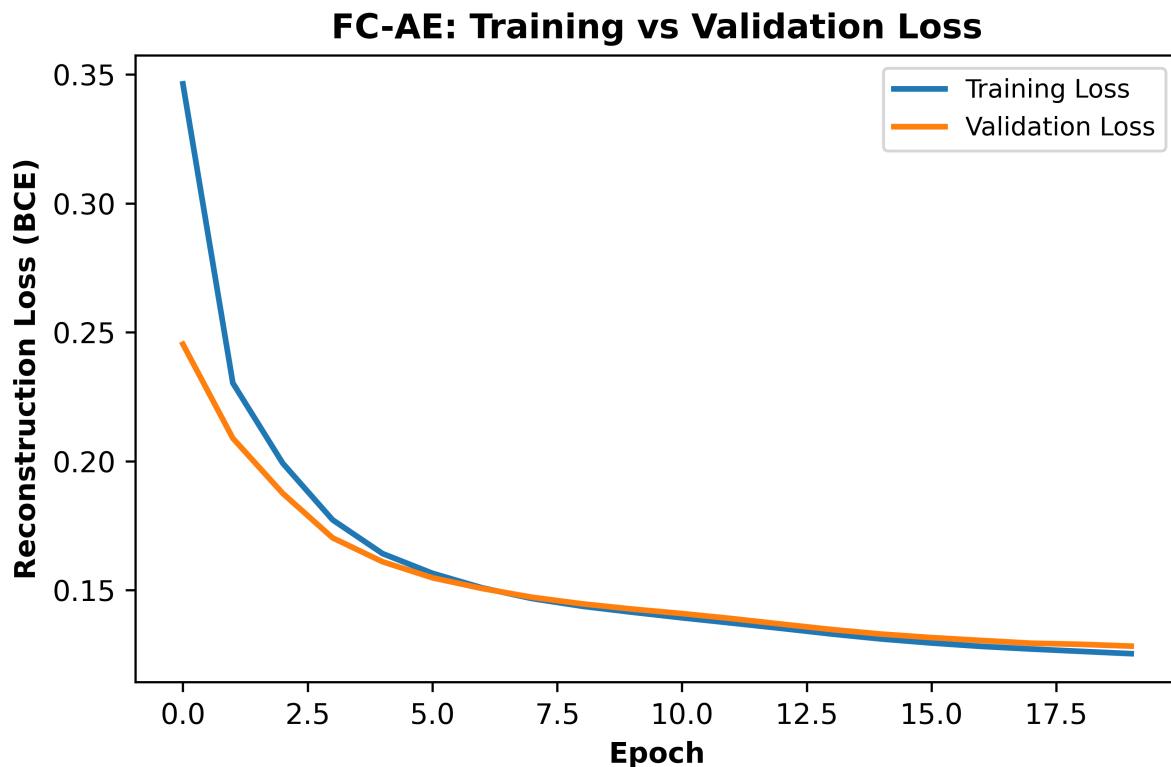


Figure 2: Training and validation reconstruction loss.

- X-axis: Epoch
- Y-axis: Reconstruction loss
- Curves: Training loss and validation loss

Inference: The model converges successfully: both the training and validation losses decrease sharply during approximately the first five epochs and then gradually approach a stable value of about 0.125 by epoch 19. This flattening indicates that the optimization process has reached a minimum and that further training would likely provide only marginal improvement. There is no clear evidence of overfitting because the training and validation loss curves remain closely aligned throughout training and converge to similar values. If overfitting were present, the validation loss would typically flatten or increase while the training loss continued to decrease.

9. Reconstruction Metrics

Students must calculate the following metrics on the test set.

Mean Squared Error

$$MSE = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2.$$

Mean Absolute Error

$$MAE = \frac{1}{N} \sum_{i=1}^N |x_i - \hat{x}_i|.$$

Structural Similarity

Calculate SSIM between the original and reconstructed images.

The implementation may use:

<https://scikit-image.org/docs/stable/api/skimetrics.html>

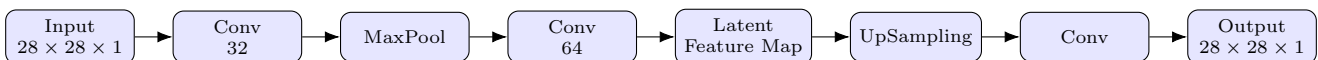
Report:

Metric	Value
Test MSE	0.00297
Test MAE	0.01628
Mean SSIM	0.9690

10. Convolutional Autoencoder

A Convolutional Autoencoder preserves spatial structure and uses convolutional operations in the encoder and upsampling/transposed convolution operations in the decoder.

The conceptual architecture is:



A possible implementation is:

```
Input(28,28,1)
Conv2D(32, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(32, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(1, 3, activation='sigmoid', padding='same')
```

The exact implementation may use Conv2DTranspose instead of UpSampling2D.

11. Comparing Fully Connected and Convolutional Autoencoders

Train both models using the same training and test images.

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.022507	0.058655	0.735063	211040
Convolutional AE	0.002974	0.016280	0.968968	74497

Plot 3: Reconstruction Comparison

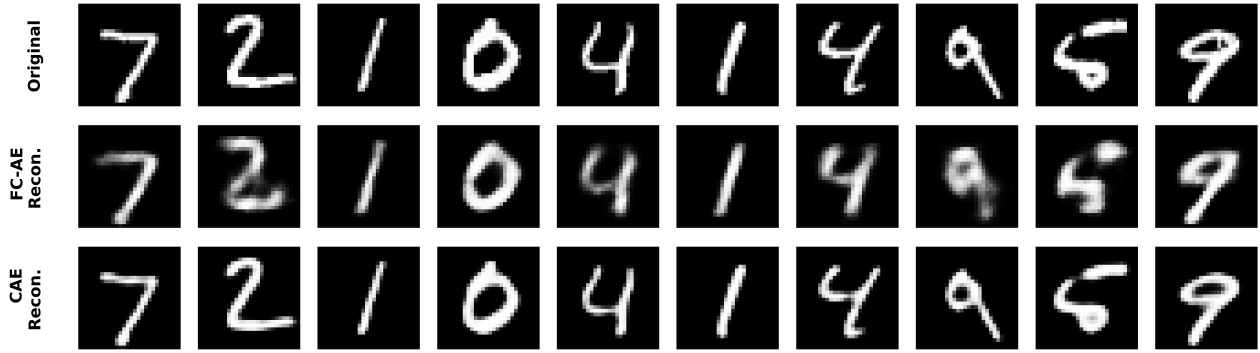


Figure 3: Comparison of the fully connected and convolutional autoencoder reconstructions.

Display:

Original $\rightarrow$ FC-AE Reconstruction $\rightarrow$ CAE Reconstruction.

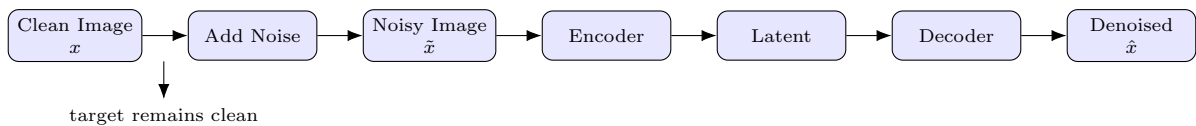
Inference: Preserving spatial locality through convolution significantly improves reconstruction performance. The CAE outperforms the FC-AE across all evaluation metrics, reducing MSE from 0.022507 to 0.002974 and MAE from 0.058655 to 0.016280, while increasing SSIM from 0.735063 to 0.968968. It achieves this performance with only 74,497 parameters, compared with 211,040 parameters for the FC-AE. Convolutional layers operate directly on local pixel neighborhoods, allowing edges, curves, loops, and other spatial patterns to be reconstructed using their surrounding context. In contrast, the FC-AE flattens each image into an unordered vector of 784 values, thereby discarding explicit spatial relationships. Moreover, weight sharing enables the CAE to learn reusable stroke-like filters instead of allocating separate weights to individual pixel connections, resulting in more accurate and parameter-efficient reconstruction.

12. Denoising Autoencoder

A denoising autoencoder receives a corrupted image but is trained to reconstruct the original clean image. The training mapping is:

$$\tilde{x} \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}$$

where $\tilde{x}$ is a noisy version of the clean image x .



13. Adding Image Noise

Use at least two corruption types.

Gaussian Noise

$$\tilde{x} = x + n, \quad n \sim \mathcal{N}(0, \sigma^2).$$

Suggested standard deviations:

$$\sigma \in \{0.1, 0.2, 0.3\}.$$

Clip the resulting image to the range $[0, 1]$.

Salt-and-Pepper Noise

Randomly replace a small proportion of pixels with values close to 0 or 1.

Suggested corruption levels:

$$p \in \{0.05, 0.10, 0.20\}.$$

14. Required Denoising Plots

Plot 4: Clean vs. Noisy vs. Denoised

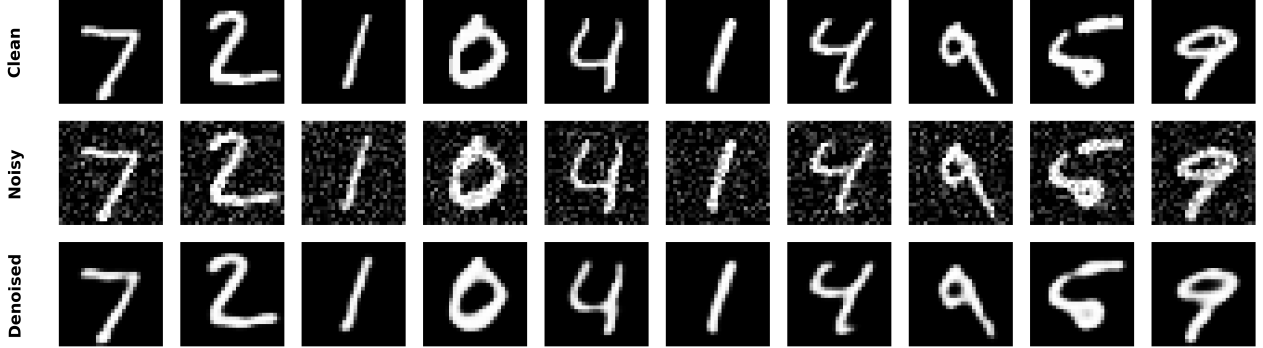


Figure 4: Clean, noisy, and denoised MNIST images.

For at least 10 images, display:

Clean | Noisy | Denoised.

Inference: Plot 4 presents the same ten digits as clean originals, noise-corrupted inputs with visible pixel-level speckling, and reconstructions produced by the convolutional denoising autoencoder. The model removes most of the visible noise and recovers strokes that closely resemble the clean images, including digits with fine details such as 1 and 4. This behavior occurs because the network is trained to map noisy inputs to clean targets. Consequently, it learns to suppress unpredictable noise while preserving the consistent and learnable structure of each digit within its latent representation. The results demonstrate that the denoising autoencoder can recover the major structure of an image even under substantial pixel corruption, indicating that its bottleneck acts as an effective structural filter rather than merely learning an identity mapping.

Plot 5: Noise Level vs. Reconstruction Quality

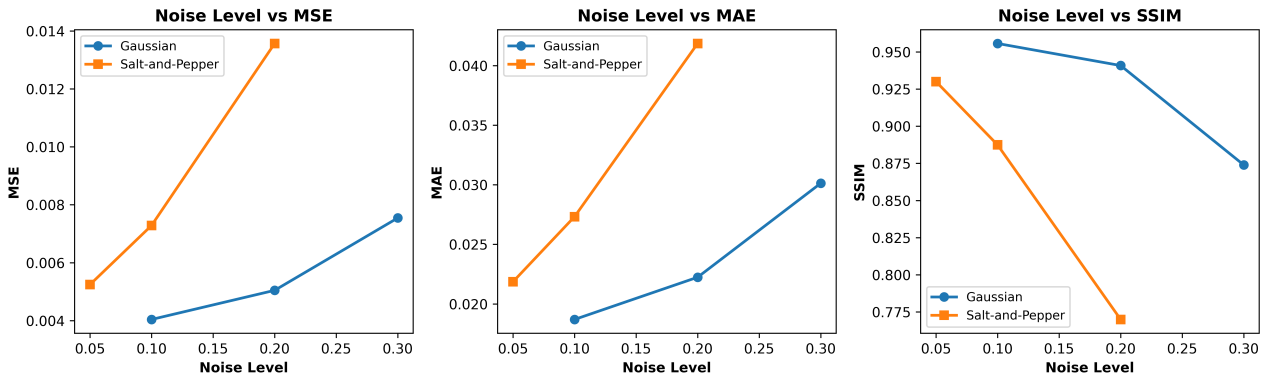


Figure 5: Effect of Gaussian noise level on reconstruction quality.

For each noise level, calculate:

MSE , MAE , $SSIM$.

Report:

Noise Level	MSE	MAE	SSIM
0.1	0.004042	0.018699	0.955667
0.2	0.005047	0.022242	0.940823
0.3	0.007544	0.030127	0.873935

Inference: Increasing the corruption level consistently reduces reconstruction quality. As the Gaussian noise standard deviation increases from 0.1 to 0.3, MSE rises monotonically from 0.004042 to 0.005047 and then 0.007544, while MAE increases from 0.018699 to 0.022242 and then 0.030127. Conversely, SSIM decreases from 0.955667 to 0.940823 and finally 0.873935. Thus, stronger corruption produces larger pixel-wise errors and weaker structural similarity. The effect is also dependent on the corruption type: salt-and-pepper noise replaces pixels with extreme values of 0 or 1, producing localized high-frequency outliers that can be more difficult for a convolutional decoder to remove than the smaller, more uniformly distributed perturbations produced by Gaussian noise.

Despite this degradation, the denoising model preserves the main structural features of the input digits. Even at the highest tested Gaussian noise level, the mean SSIM remains above 0.87. The clean-noisy-denoised comparison in Plot 4 also shows that the principal digit shapes are recovered despite the corruption. The gradual rather than abrupt decline in performance suggests that the model has learned a meaningful mapping from corrupted inputs to the underlying clean-digit manifold instead of merely memorizing the training samples.

15. Variational Autoencoder

A conventional autoencoder maps an input to a deterministic latent vector:

$$z = f(x).$$

A VAE instead learns a probability distribution:

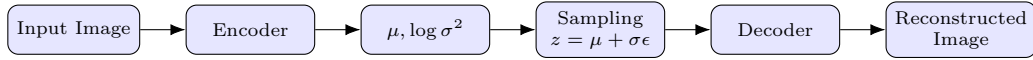
$$q_\phi(z|x) = \mathcal{N}(\mu(x), \text{diag}(\sigma^2(x))).$$

The encoder produces:

$$\mu, \quad \log \sigma^2.$$

A latent vector is sampled using the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$



16. VAE Loss Function

The VAE objective consists of two components.

Reconstruction Loss

$$\mathcal{L}_{rec} = -\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)].$$

For normalized MNIST images, binary cross-entropy can be used.

KL Divergence

$$D_{KL}(q_\phi(z|x) \parallel p(z)) = -\frac{1}{2} \sum_j (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2).$$

The total loss is

$$\boxed{\mathcal{L}_{VAE} = \mathcal{L}_{rec} + \mathcal{L}_{KL}}$$

where the prior is usually

$$p(z) = \mathcal{N}(0, I).$$

17. VAE Implementation

Use a small latent space so that it can be visualized.

Suggested latent dimension:

$$z \in \mathbb{R}^2.$$

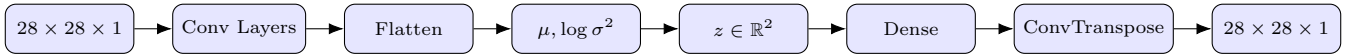
The encoder should produce:

$$\mu \in \mathbb{R}^2, \quad \log \sigma^2 \in \mathbb{R}^2.$$

The decoder maps

$$z \rightarrow 28 \times 28 \times 1.$$

A compact architecture is:



18. Exploring the VAE Latent Space

Since the latent dimension is two, students can visualize the encoded test images in a two-dimensional plane.

Plot 6: VAE Latent Space

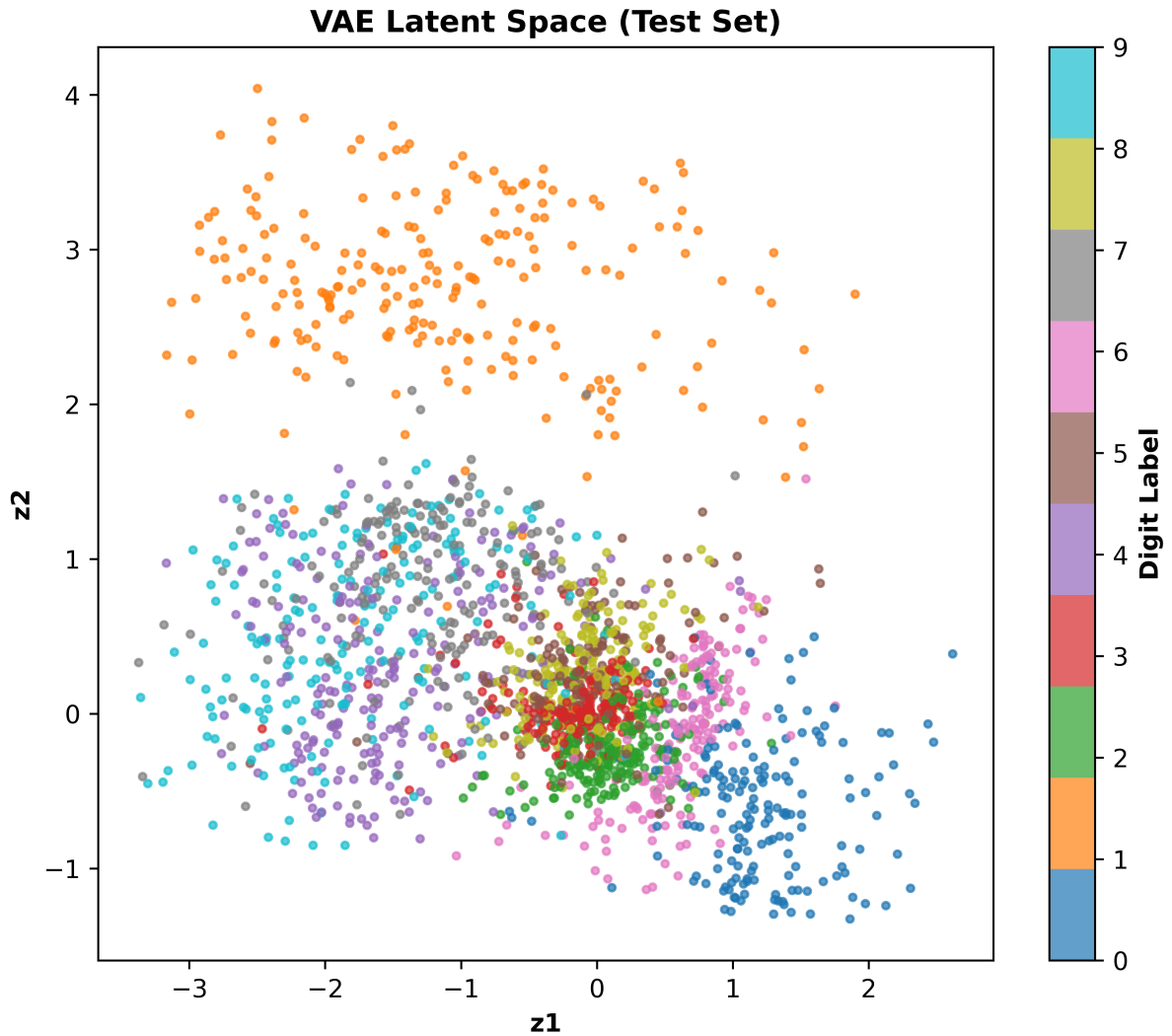


Figure 6: Two-dimensional VAE latent-space visualization.

- X-axis: z_1
- Y-axis: z_2
- Color/marker category: digit label

The labels are used only for visualization; they are not supplied to the VAE during training.

Inference: Samples belonging to the same digit generally occupy similar regions of the latent space. In particular, digit 1 forms a compact, well-separated cluster in the upper region of the plot, whereas digits such as 0 form more dispersed local clusters in the lower region. However, substantial overlap exists among different digit classes, with many of the remaining classes crowded together without clear boundaries. This overlap is expected because the VAE is trained without class labels and compresses the visual variation of ten digit classes into only two latent dimensions. Thus, the latent space captures broad similarities among digit shapes but does not provide complete class separation.

19. Generating New Images with the VAE

Sample points from

$$z \sim \mathcal{N}(0, I).$$

Pass the sampled latent vectors through the decoder.

Plot 7: Randomly Generated Images

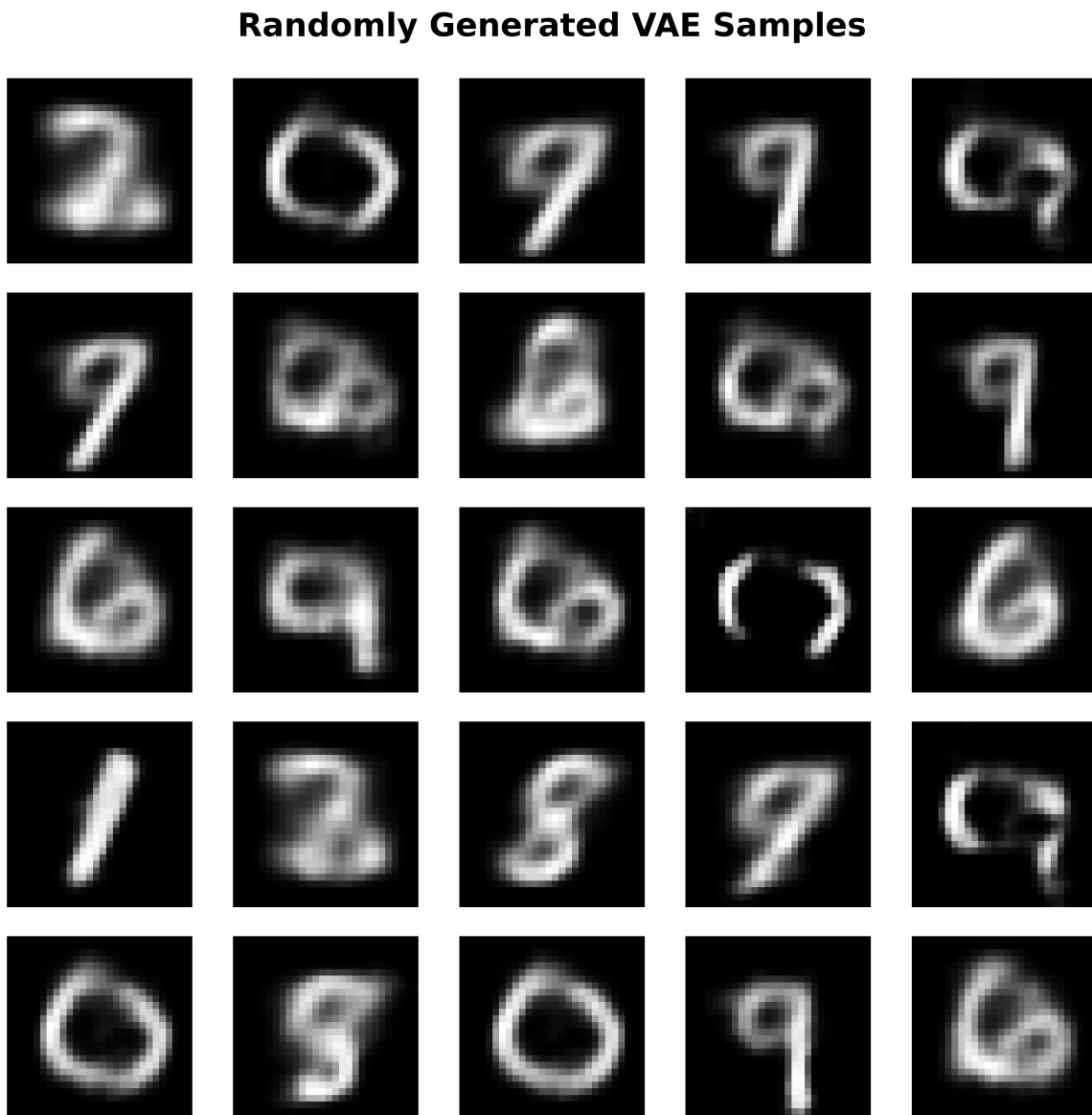


Figure 7: Images generated by sampling from the VAE latent distribution.

Generate at least 25 images using randomly sampled latent vectors.

Display them in a grid.

Students should determine:

- whether the generated images resemble handwritten digits;
- whether digit shapes are recognizable;
- whether there is visible variation among generated samples; and
- whether some samples are ambiguous or unrealistic.

Inference: Most of the 25 generated samples resemble handwritten digits, with shapes similar to 0, 6, and 9 appearing frequently. The majority are identifiable, demonstrating that the VAE has learned meaningful features of the MNIST data distribution. However, some samples are unclear or unrealistic: a few combine characteristics of multiple digits, such as a 6/8-like shape, while others resemble non-digit forms such as brackets. These ambiguous outputs can occur when a vector sampled from $\mathcal{N}(0, I)$ lies between digit-class regions or in a sparsely represented part of the learned latent space. Overall, the VAE successfully generates diverse digit-like samples, although their visual quality and clarity vary across the sampled points.

20. Latent Space Interpolation

Select two latent vectors:

$$z_A, \quad z_B.$$

Construct

$$z(\alpha) = (1 - \alpha)z_A + \alpha z_B, \quad \alpha \in [0, 1].$$

Use values such as:

$$\alpha = 0, 0.1, 0.2, \dots, 1.$$

Decode every interpolated vector.

Plot 8: Latent Space Interpolation

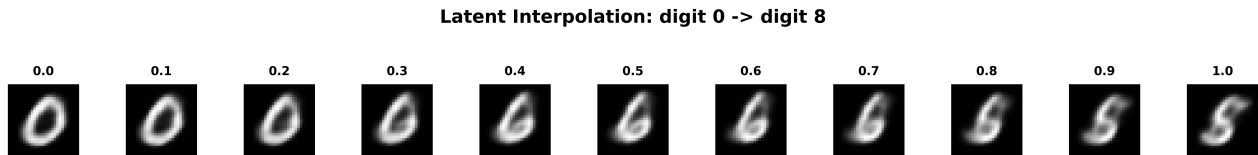


Figure 8: Decoded images along a linear interpolation in the VAE latent space.

Display the decoded images in sequence.

Inference: The decoded image changes continuously as the latent vector moves from the representation of 0 to that of 8. No sudden transformation occurs; instead, the rounded shape of 0 gradually develops the additional strokes required to resemble 8 as α increases from 0 to 1. This smooth transition is enabled by the continuous VAE latent space, which is regularized toward the standard normal distribution $\mathcal{N}(0, I)$. Consequently, neighboring points tend to decode into visually similar images, allowing meaningful intermediate digit forms to appear along the interpolation path.

21. Reconstruction Comparison Across Models

Compare the reconstruction performance of all four models:

Model	MSE	MAE	SSIM	Parameters
FC Autoencoder	0.022507	0.058655	0.735063	211040
Convolutional AE	0.002974	0.016280	0.968968	74497
Denoising CAE	0.005036	0.022221	0.940905	74497
VAE	0.044173	0.102887	0.483118	284933

The VAE loss components are reported separately below because its total objective also contains the KL-divergence term.

VAE Metric	Value
Reconstruction Loss	149.843140
KL Loss	6.049046
Total Loss	155.892181
Test MSE	0.044173
Test MAE	0.102887
Mean SSIM	0.483118

22. Required Plots

The final report must contain the following plots:

1. Original image versus reconstructed image.
2. Training and validation loss for the basic autoencoder.
3. Fully connected AE versus convolutional AE reconstruction.
4. Clean versus noisy versus denoised images.
5. Noise level versus MSE/MAE/SSIM.
6. VAE latent-space visualization.
7. Randomly generated VAE images.
8. Latent-space interpolation.
9. Training/validation reconstruction loss for the VAE.
10. Reconstruction-error distribution on the test set.

For plots containing multiple quantitative measures, use separate figures or clearly labeled axes rather than placing incompatible scales on one axis.

23. Reconstruction Error Distribution

Calculate the per-image reconstruction error:

$$e_i = \frac{1}{D} \sum_{j=1}^D (x_{ij} - \hat{x}_{ij})^2.$$

Plot 9: Distribution of Reconstruction Errors

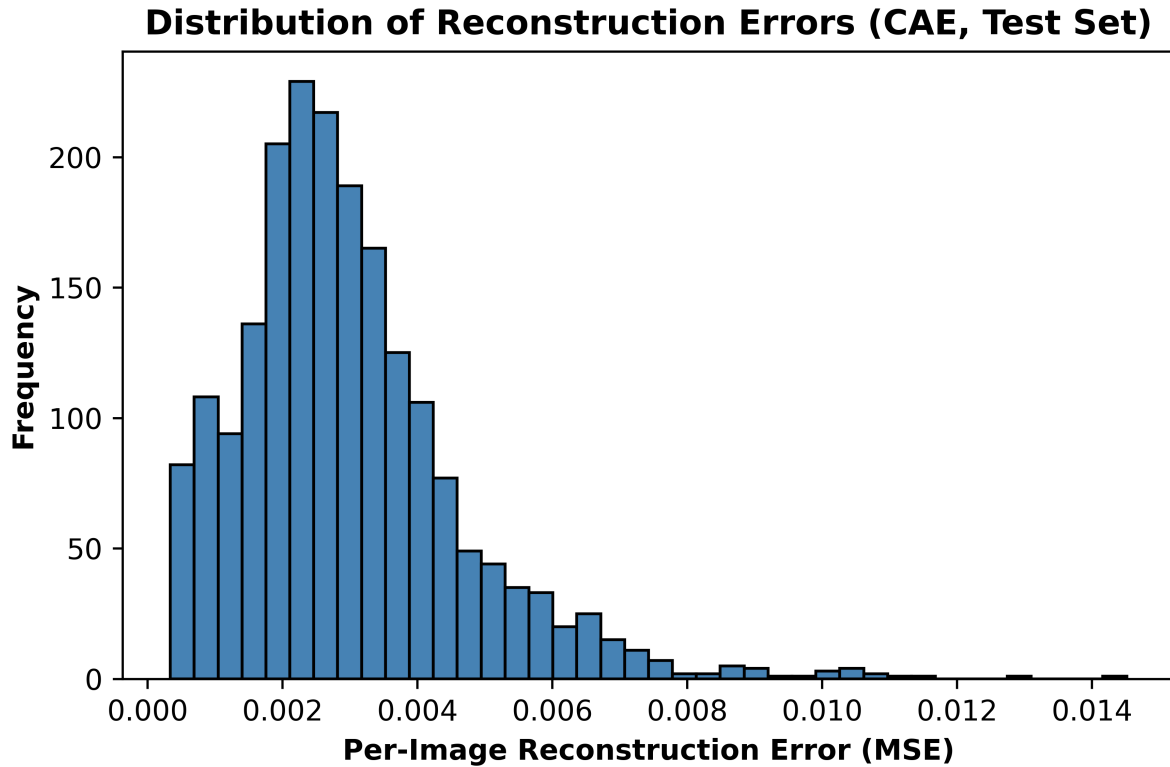


Figure 9: Distribution of per-image reconstruction errors on the test set.

Plot a histogram of e_i over the test set.

Students should identify:

- the typical reconstruction-error range;
- whether the distribution is concentrated or spread out;
- possible high-error samples; and
- representative images corresponding to high reconstruction error.

Inference: Most test images have reconstruction errors in the range of approximately 0.001–0.004 MSE. The distribution is highly concentrated and right-skewed rather than broadly scattered, with a thin tail extending to approximately 0.014–0.015. This tail contains only a small number of high-error outliers. Such samples likely contain unusual handwriting styles or structures that are poorly represented by the image manifold learned by the autoencoder. This interpretation is supported by the five largest errors, which range from 0.010840 to 0.014516 MSE and are substantially higher than the typical reconstruction-error range.

24. Required High-Error Image Analysis

Select the five test images having the largest reconstruction errors.

Display:

Sample	Reconstruction
1	0.014516
2	0.012787
3	0.011427
4	0.011013
5	0.010840

Students should provide a short explanation for why these images may be difficult to reconstruct.

25. Consolidated Results

Model	MSE	MAE	SSIM	Parameters	Time (s)
FC Autoencoder	0.022507	0.058655	0.735063	211040	18.790881
Convolutional AE	0.002974	0.016280	0.968968	74497	19.484384
Denoising CAE	0.005036	0.022221	0.940905	74497	16.569801
VAE	0.044173	0.102887	0.483118	284933	29.158941

The values must be obtained from the student's own execution.

26. Required Inferences

For every major plot, students must write a short inference containing:

1. What does the plot show?
2. What trend is observed?
3. Why might the observed trend occur?
4. What conclusion can be drawn?

The following inferences are mandatory:

- effect of latent dimension on reconstruction;
- difference between fully connected and convolutional reconstruction;
- effect of Gaussian noise level;
- ability of the denoising autoencoder to recover image structure;
- characteristics of the VAE latent distribution;
- structure of the two-dimensional VAE latent space;
- quality and diversity of generated samples;
- smoothness of latent-space interpolation;
- relationship between reconstruction error and visual quality; and
- differences between deterministic and probabilistic latent representations.

27. Additional Latent-Dimension Study

Repeat the basic autoencoder using:

$$d_z \in \{2, 8, 16, 32\}.$$

Report:

Latent Channels	MSE	MAE	SSIM
4	0.019347	0.053206	0.792876
8	0.010580	0.034231	0.893574
16	0.005868	0.023523	0.940982
32	0.004461	0.020037	0.954720

Plot 10: Latent Dimension vs. Reconstruction MSE

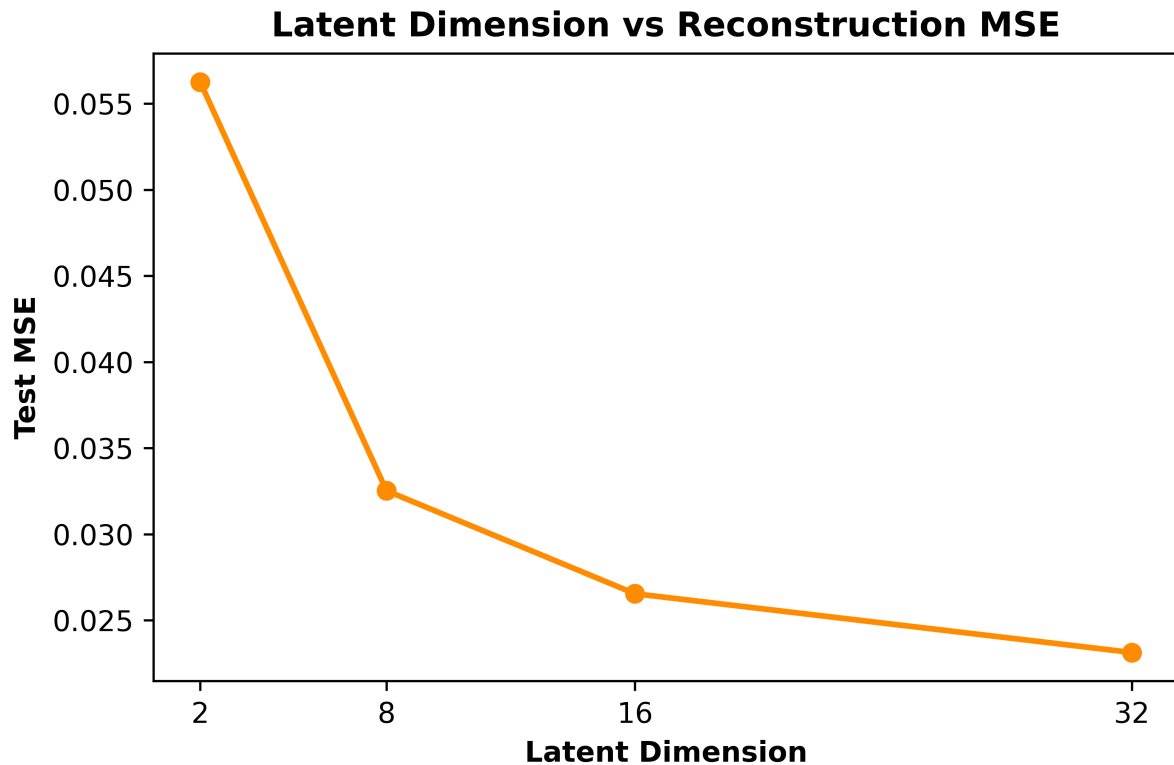


Figure 10: Effect of latent-channel capacity on reconstruction MSE.

Inference: The results demonstrate a clear trade-off between compression and reconstruction quality. Using only 4 latent channels creates a strong bottleneck and produces the highest MSE of 0.019347. Increasing the latent capacity to 8, 16, and 32 channels allows the model to preserve more image information, reducing MSE to 0.010580, 0.005868, and 0.004461, respectively. The largest improvements occur at the lower capacities, whereas the curve begins to level off beyond 16 channels. This diminishing return suggests that the latent representation has already captured most of the intrinsic visual structure of the MNIST digits, so additional channels provide only modest improvements in reconstruction quality.

28. Discussion Questions

1. What is an autoencoder? ans: auto encoder is an neural network that compresses and input image to a single layered bottle and then automatically reconstructs the image as close as the input that it learning from the compressed code
2. What is the purpose of the encoder? ans: it take the input which is a raw data and extract features and patterns and compress it into a condensed context vectore
3. What is the purpose of the decoder? ans: it take the compressed context vector and uses it reconstruct it into a human reabddable format or an targeted output
4. What is a latent representation? ans: a compressed encoded feature map that captures the most salient features
5. Why is the latent representation usually smaller than the input? ans: a smaller latent space forces the network to learn compact informative encoding rather than trivial and useless information
6. What happens when the latent dimension is too small? ans: the bottleneck will not be able to retain information to reconstruct the target or the input image correctly
7. What happens when the latent dimension is increased? ans: the increase in latent dimesnion improves the reconstruction quality as it can store more information
8. Why can a convolutional autoencoder preserve spatial information more effectively? Convs use shared weights over local pixel neighborhoods to preserve spatial layout, whereas fully connected layers flatten images and lose

spatial context.

9. Differentiate an autoencoder and a denoising autoencoder. ans: a plain AE is taught with input and target and denoising AE is taught iwth corrupted input and clean target output, which forces it to become more robust to noise
10. Why is the clean image used as the target in denoising? ans: to teach the model that the objective is to recover the true underlying signal from the corrupted observation
11. What happens when the noise level is increased? ans: reconstruction quality degrades and the evaluton metrices values move towards degradtion and the original input may get destroyed in this process
12. Why are MSE, MAE and SSIM useful for image reconstruction? ans: MSE AND MAE quantify average pixel wise intensity error, SSIM captures structural similairty that pizel wise metric can miss
13. What is a Variational Autoencoder? ans: is the generative variant autoencoder that learns the probality distf-bution over the latent space onsteast of single latent vector
14. How does a VAE differ from a conventional autoencoder? a conventional autoencoder learns deterministic mapping while the VAE learns the parameters over the distribution, which helps in generation
15. Why does a VAE learn a distribution instead of a single latent vector? ans: Learning a distributionmakes the latent space continuous and well-structured, so that any point sampled from the prior decodes into a plausible output
16. What is the reparameterization trick? ans: rewriting the stochastic step so that gradients can backpropagate through mean and variance depsite the sampling step
17. Why is ϵ sampled from a standard normal distribution? ans: it isolates the randomness into a variable that carries no trainable parameters
18. What is the role of KL divergence in the VAE loss? ans: it measure how far the latent distribution is the from the prior knowlege and also acts like a regulaizer pill the images encoded distribution towards a shared prior
19. What is the prior distribution used in the VAE? ans: The prior is a standard multivariate normal distribution, $p(z) = \mathcal{N}(0, I)$.
20. Why is a two-dimensional latent space useful for visualization? ans: it can be directly plotted on the scatter plot and the class structure can learn the representation visually
21. What does latent-space interpolation demonstrate? ans: shows that the latent space is smothh and continuous as z takes small steps instead of jumping to noise
22. Why can a VAE generate new images? ans: VAE forces its latent space into an organized standard normal distribution, allowing us to pick any random point z and decode it into a new, realistic image.
23. Why might some VAE-generated images be unclear? ans: the images will turn out unclear as the decoder gets confused when generating outputs due to empty spaces and overlapping boundaries
24. What is the difference between reconstruction and generation? ans: reconstruction tries to reproduce the given input as such with littlw to no changes, generation starts from the input vector and tries to produce an entirely new sample
25. Why should test images remain unseen during training? ans: to obtain and unbiansed evaluation of the model and see how well it can generalize on unseen data

29. Additional Exercises

1. Train the convolutional autoencoder with latent dimensions of 4, 8, 16 and 32 and compare reconstruction quality.

Latent Channels	MSE	MAE	SSIM
4	0.019347	0.053206	0.792876
8	0.010580	0.034231	0.893574
16	0.005868	0.023523	0.940982
32	0.004461	0.020037	0.954720

2. Compare Gaussian noise and salt-and-pepper noise using the same denoising architecture.

Noise Level	MSE	MAE	SSIM	Type
0.10	0.004042	0.018699	0.955667	Gaussian
0.20	0.005047	0.022242	0.940823	Gaussian
0.30	0.007544	0.030127	0.873935	Gaussian
0.05	0.005245	0.021863	0.930027	Salt-and-Pepper
0.10	0.007285	0.027324	0.887436	Salt-and-Pepper
0.20	0.013567	0.041856	0.769977	Salt-and-Pepper

3. Train the denoising autoencoder using two different noise levels and compare SSIM.

Training Noise Level	MSE	MAE	SSIM
0.1	0.004105	0.019418	0.955713
0.3	0.008128	0.029516	0.901971

4. Replace the convolutional decoder with transposed convolutions and compare the reconstructions.

Decoder	MSE	MAE	SSIM
UpSampling2D (Section 10 CAE)	0.002974	0.016280	0.968968
Conv2DTranspose	0.000835	0.008888	0.990237

5. Implement a VAE with latent dimension 2 and 8 and compare the latent representations.

Latent Dimension	MSE	MAE	SSIM
2	0.046044	0.107249	0.454150
8	0.023984	0.061238	0.734471

6. Investigate the effect of increasing the KL-loss contribution.

β (KL Weight)	MSE	SSIM
0.5	0.049738	0.425492
1.0	0.046994	0.437808
5.0	0.050113	0.417584

7. Generate a larger grid of VAE samples and discuss diversity.

100 Randomly Generated VAE Samples

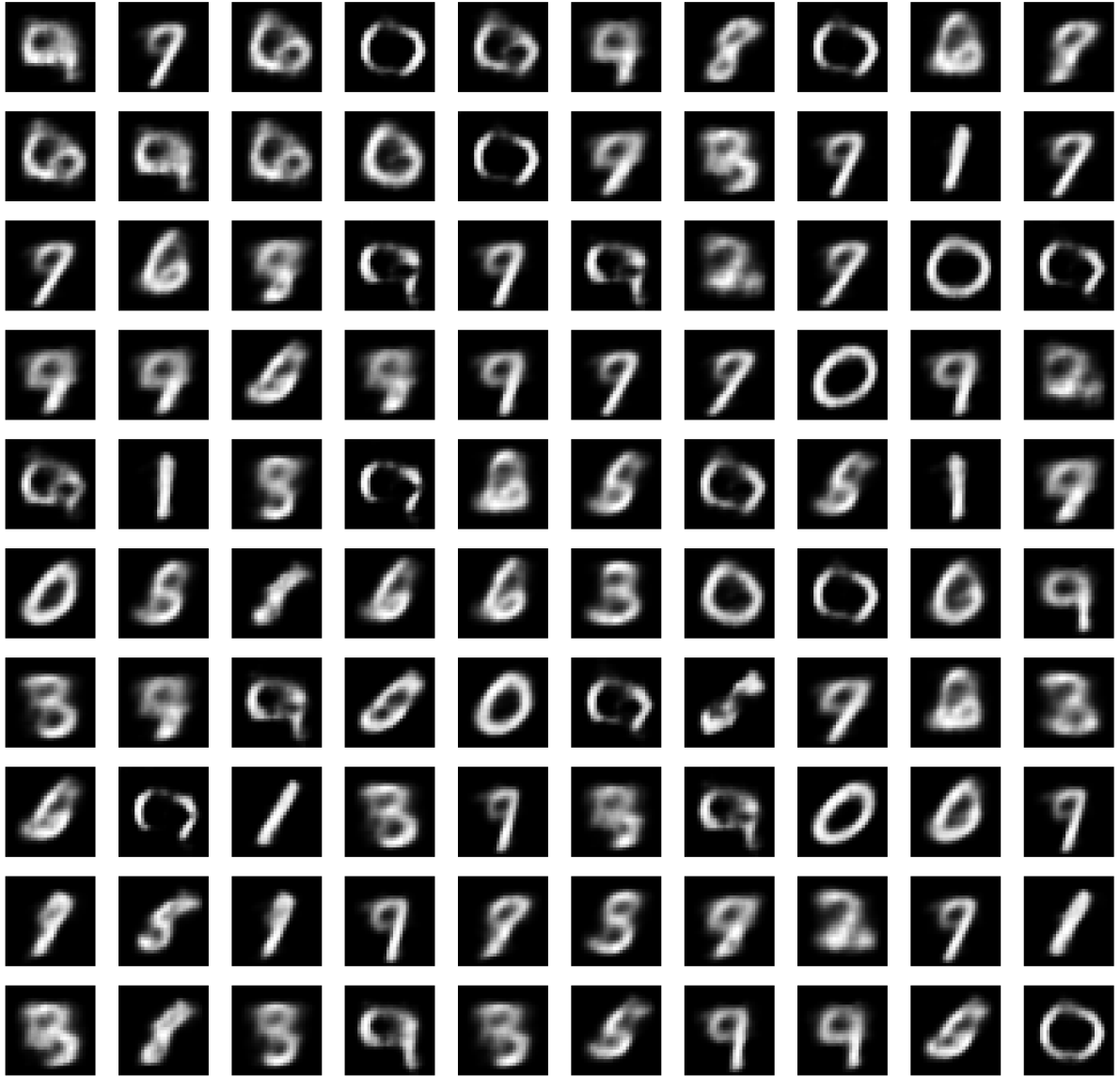


Figure 11: A 10×10 grid of randomly generated VAE samples.

Inference: The 10×10 grid of 100 randomly generated samples exhibits greater visible diversity than the earlier grid of 25 samples. The images vary considerably in stroke thickness, stroke angle, and overall digit shape, while digits containing loops, such as 0, 6, 8, and 9, appear more frequently than simpler digits such as 1. The larger sample size also makes the strengths and limitations of the model more apparent: the VAE can produce many clear and recognizable digits, but it occasionally generates ambiguous or mixed forms. This confirms that the unclear samples observed in the smaller grid are not isolated cases; they reflect overlap among digit regions in the two-dimensional latent space, as also indicated by Plot 6.

8. Select two digit regions in the latent space and perform interpolation.

Interpolation Between Digit-1 and Digit-7 Cluster Centroids



Figure 12: Latent-space interpolation between the centroids of digits 1 and 7.

Inference: Rather than interpolating between two arbitrary samples, this experiment interpolates between the mean latent positions, or centroids, of all digit-1 and digit-7 test samples. Because each centroid represents the center of its corresponding digit cluster, the resulting path describes a more representative transformation from a typical 1 to a typical 7 and is less affected by unusual individual handwriting. The decoded sequence changes

smoothly from a primarily vertical stroke into the horizontal-and-diagonal structure of a 7. The intermediate images resemble plausible transitional pen strokes rather than abrupt or unrealistic jumps. This result demonstrates that the VAE latent space is smooth not only between individual samples but also between broader class regions.

30. Expected Outcome

At the end of the experiment, students should be able to demonstrate the complete image-representation and reconstruction pipeline:

Image $\rightarrow$ Encoder $\rightarrow$ Latent Representation $\rightarrow$ Decoder $\rightarrow$ Reconstructed Image

They should also demonstrate denoising:

Clean Image $\rightarrow$ Noise $\rightarrow$ Denoising Autoencoder $\rightarrow$ Recovered Image

and variational generation:

Image $\rightarrow (\mu, \sigma) \rightarrow z \rightarrow$ Decoder $\rightarrow$ Generated/Reconstructed Image

The final report should contain both quantitative evaluation and visual analysis. Numerical performance values must be obtained from the student's own execution.

31. References

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